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A Robust Planning Model for Offshore Microgrid Considering Tidal Power and Seawater Desalination

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01

Background



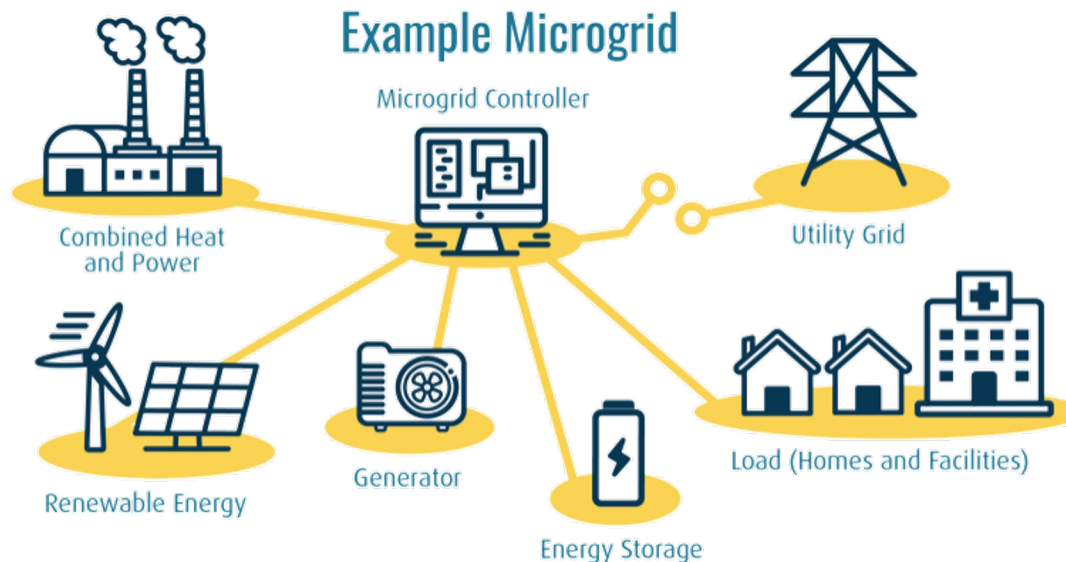
Microgrid

Definition by US Department of Energy:

Microgrid is “a group of interconnected loads and **distributed energy resources** within clearly defined electrical boundaries that acts as a single controllable entity with respect to the grid. A microgrid can connect and disconnect from the grid to enable it to operate in both **grid-connected** or **island-mode**.”

Advantages of Microgrid [1]:

- Improve power quality.
- Reduce emissions.
- Reduce network congestion and power losses.
- Increase energy efficiency.
- Improve system economics.
- The ability of islanding in the event of faults or disturbances in upstream networks enhances grid and customers' reliability and resilience.



Offshore Microgrid



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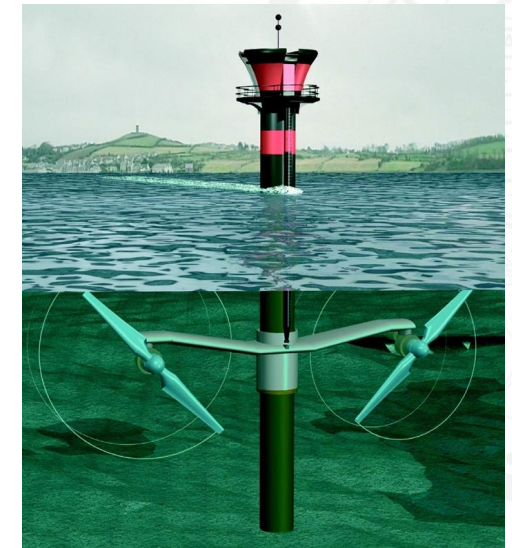
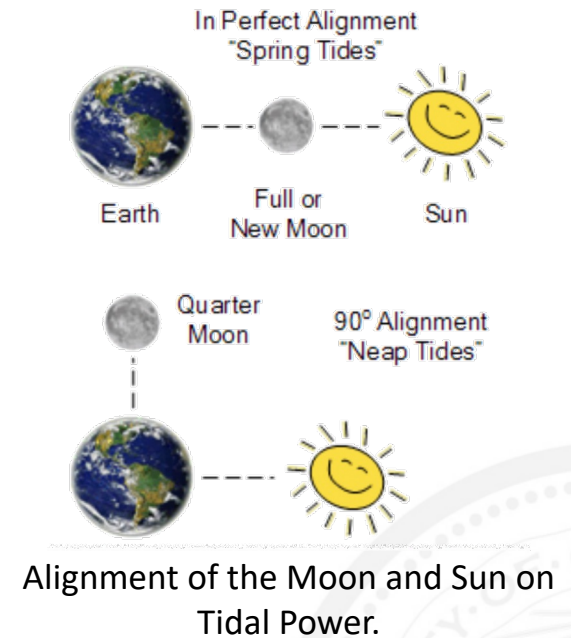
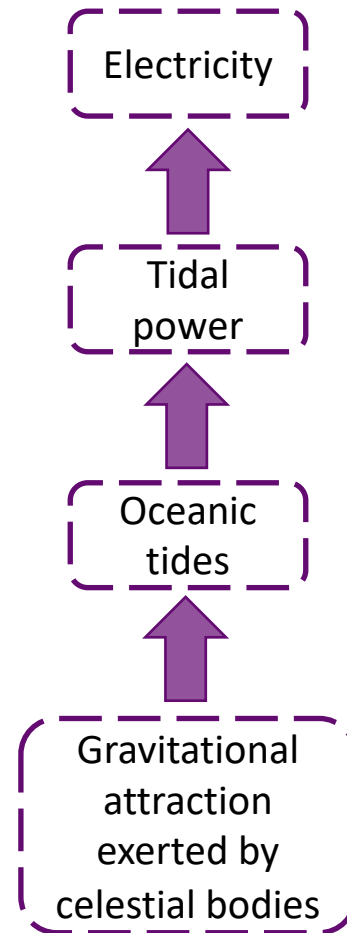
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- Abundant resources of **ocean renewable energy**.
- Higher probability and frequency of **natural disasters**, which are more likely to lead to damages and power outages.

Tidal Power Generation

- Tidal power is taken from the Earth's oceanic tides.
- Tidal forces result from periodic variations in gravitational attraction exerted by celestial bodies.
- The forces create corresponding motions in the world's oceans, resulting in periodic changes in sea levels.
- These changes are highly regular and predictable, due to the consistent pattern of the Earth's rotation and the Moon's orbit around the Earth.
- A tidal generator converts the energy of tidal flows into electricity. Greater tidal variation and higher tidal current velocities can dramatically increase the potential of a site for tidal electricity generation.



A typical tidal generator.



02

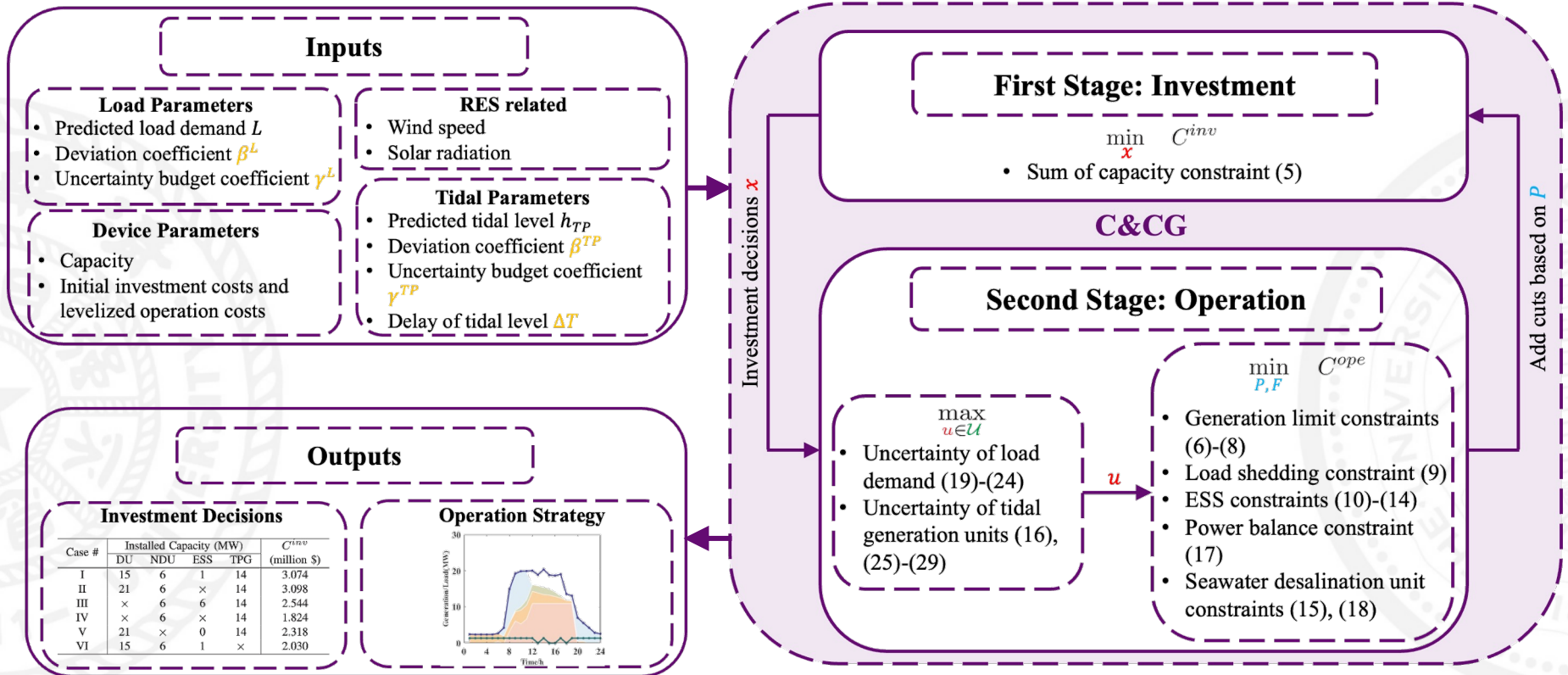
Model Formulation



Model Formulation

Framework of the Planning Problem

Red letters: binary decision variables.
Blue letters: continuous decision variables.
Green letters: uncertainty sets.
Yellow letters: uncertainty-related parameters.



Model Formulation

Robust Planning Model: Objective Function

$$\begin{aligned}
 \min_x C^{inv} &\square \max_u \min_{P,LS,F} C^{ope} \\
 C^{inv} &\square \sum_y \kappa_y \left[\begin{aligned} &\sum_{i \in \Omega_{DU}} cc_i RC_i x_i \\ &\square \sum_{j \in \Omega_{NDU}} cc_j RC_j x_j \\ &\square \sum_{k \in \Omega_{TPG}} cc_k RC_k x_k \\ &\square \sum_{l \in \Omega_{ESS}} \square cp_l RP_l \square ce_l RC_l \square x_l \\ &\square cc_F RC_F \end{aligned} \right] \\
 C^{ope} &\square \sum_y \sum_d \sum_h \left[\begin{aligned} &\sum_{i \in \Omega_{DU}} c_i P_{i,h,d,y} \\ &\square \sum_{k \in \Omega_{TPG}} c_k P_{k,h,d,y} \\ &\square c_F F_{h,d,y} \square v LS_{h,d,y} \end{aligned} \right], \forall h, d, y
 \end{aligned}$$



Model Formulation

Deterministic Planning Model: Constraints

$$D^{\max} \leq \sum_i RP_i x_i \leq \sum_j RP_j x_j \leq \sum_k RP_k x_k$$

$$0 \leq P_{i,h,d,y} \leq RP_i x_i$$

$$0 \leq P_{j,h,d,y} \leq RP_j x_j$$

$$0 \leq P_{k,h,d,y} \leq RP_k x_k$$

$$0 \leq LS_{h,d,y} \leq D_{h,d,y}$$

$$0 \leq P_{l,h,d,y}^{dch} \leq RP_l x_l$$

$$0 \leq P_{l,h,d,y}^{ch} \leq RP_l x_l$$

$$SOC_{l,h,d,y} \leq SOC_{l,h,d,y} \leq P_{l,h,d,y}^{ch} \eta_l^{ESS} - P_{l,h,d,y}^{dch} / \eta_l^{ESS}$$

$$0 \leq SOC_{l,h,d,y} \leq RC_l x_l$$

$$SOC_{l,1,d,y} \leq SOC_{l,24,d,y}$$

$$0 \leq F_{h,d,y} \leq RC_F$$

$$\sum_i P_{i,h,d,y} \leq \sum_j P_{j,h,d,y} \leq \sum_k P_{k,h,d,y} \leq \sum_l [P_{l,h,d,y}^{dch} - P_{l,h,d,y}^{ch}] \leq D_{h,d,y} \leq \alpha_F F_{h,d,y} - LS_{h,d,y}$$

$$\sum_h F_{h,d,y} \geq F_0$$



Model Formulation

Uncertainty Characterization: Load Demand

$$\mathcal{U}_D : D_{h,d,y} \in [\tilde{D}_{h,d,y} - \underline{D}_{h,d,y}, \tilde{D}_{h,d,y} + \bar{D}_{h,d,y}]$$

$$\underline{u}_{h,d,y}^D \in [\underline{u}_{h,d,y}^D, \bar{u}_{h,d,y}^D] \leq 1$$

$$\bar{D}_{h,d,y} \in [\underline{D}_{h,d,y}, \beta_D \tilde{D}_{h,d,y}]$$

$$\sum_d \sum_h [\underline{u}_{h,d,y}^D \in [\underline{u}_{h,d,y}^D, \bar{u}_{h,d,y}^D]] \leq \Gamma_y^D$$

$$\Gamma_y^D \in [\gamma_y^D \cdot D \cdot H] \quad \text{Uncertainty budget}$$

$$\forall h, d, y$$

Model Formulation

Uncertainty Characterization: Tidal Power Generation

$$\mathcal{U}_{TPG} : P_{k,h,d,y}^{TPG} \sqsubseteq \tilde{P}_{k,h,d,y}^{TPG} - \underline{P}_{k,h,d,y}^{TPG} \underline{u}_{k,h,d,y}^{TPG} \sqsubseteq \bar{P}_{k,h,d,y}^{TPG} \bar{u}_{k,h,d,y}^{TPG}$$

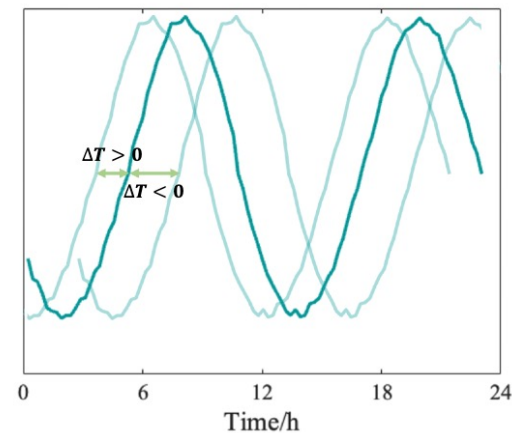
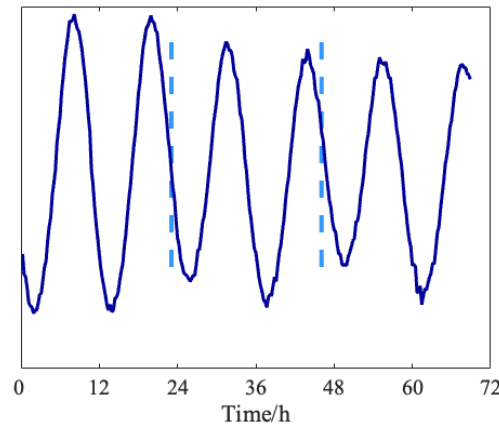
$$\underline{u}_{k,h,d,y}^{TPG} \sqsubseteq \bar{u}_{k,h,d,y}^{TPG} \leq 1$$

$$\bar{P}_{k,h,d,y}^{TPG} \sqsubseteq \underline{P}_{k,h,d,y}^{TPG} \sqsubseteq \beta_{TPG} \tilde{P}_{k,h,d,y}^{TPG}$$

$$\sum_d \sum_h \left[\underline{u}_{k,h,d,y}^{TPG} \sqsubseteq \bar{u}_{k,h,d,y}^{TPG} \right] \leq \Gamma_y^{TPG}$$

$$\Gamma_y^{TPG} \sqsubseteq \gamma_y^{TPG} \cdot D \cdot H$$

$$\forall k \in \Omega_{TPG}, \forall h, d, y$$





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Case Studies



Case Studies

The Discussed Offshore Microgrid

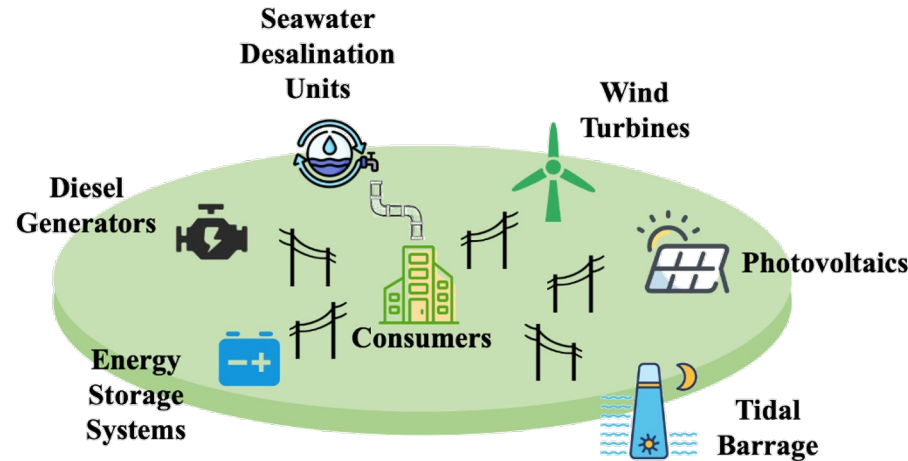


TABLE I
PARAMETERS OF THE CANDIDATE DUS

Unit No.	Capacity (MW)	Levelized Operation Cost (\$/MWh)	Initial Investment Cost (\$/MW)
1	6	140	44,000
2	5	130	54,000
3	4	120	64,000
4	3	110	74,000
5	2	100	84,000
6	1	90	94,000

TABLE II
PARAMETERS OF THE CANDIDATE NDUS

Unit No.	Capacity (MW)	Levelized Operation Cost (\$/MWh)	Initial Investment Cost (\$/MW)	Type of Energy
1	4	-	150,000	WT
2	2	-	90,000	PV

TABLE III
PARAMETERS OF THE CANDIDATE ESSs

Unit No.	Rated Power (MW)	Rated Energy (MWh)	Initial Investment Cost - Power (\$/MW)	Initial Investment Cost - Energy (\$/MWh)
1	1	6	60,000	30,000
2	2	6	30,000	30,000
3	3	6	20,000	30,000

TABLE IV
PARAMETERS OF THE CANDIDATE TPG UNITS

Unit No.	Capacity (MW)	Levelized Operation Cost (\$/MWh)	Initial Investment Cost (\$/MW)
1	5	-	54,000
2	4	-	72,000
3	3	-	90,000
4	2	-	108,000



Case Studies

Effect of Robustness

COMPARISON OF INVESTMENT DECISIONS AND COSTS
BETWEEN RPM AND DPM UNDER DIFFERENT DER CONFIGURATIONS

Case #		Installed Capacity (MW)				C^{inv} (M \$)	LS (MW)
		DU	NDU	ESS	TPG		
I	DPM	15	6	5	14	3.314	37.47
	RPM	21	6	5	14	3.578	
II	DPM	15	6	-	14	2.834	49.31
	RPM	21	6	-	14	3.098	
V	DPM	15	-	6	14	2.774	66.04
	RPM	21	-	6	14	3.038	
VI	DPM	15	6	3	×	2.030	61.51
	RPM	21	6	5	-	2.534	

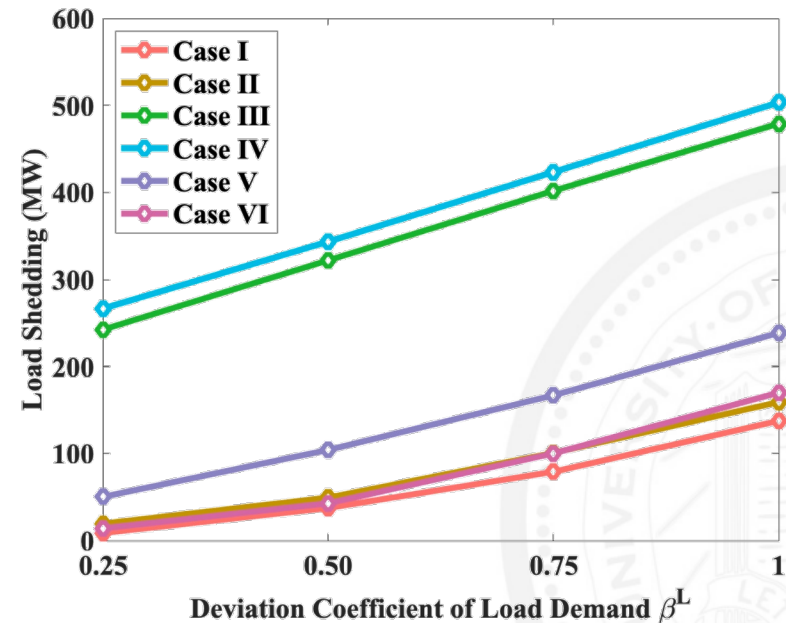
- Load curtailment when implementing investment decisions from DPM into RPM, demonstrating the robustness of the RPM.
- More DU invested when taking uncertainty into consideration.
- Higher investment costs of RPM.

Case Studies

Analysis of Uncertainty of Load Demand

INVESTMENT DECISIONS AND COSTS OF DPM UNDER DIFFERENT DER CONFIGURATIONS

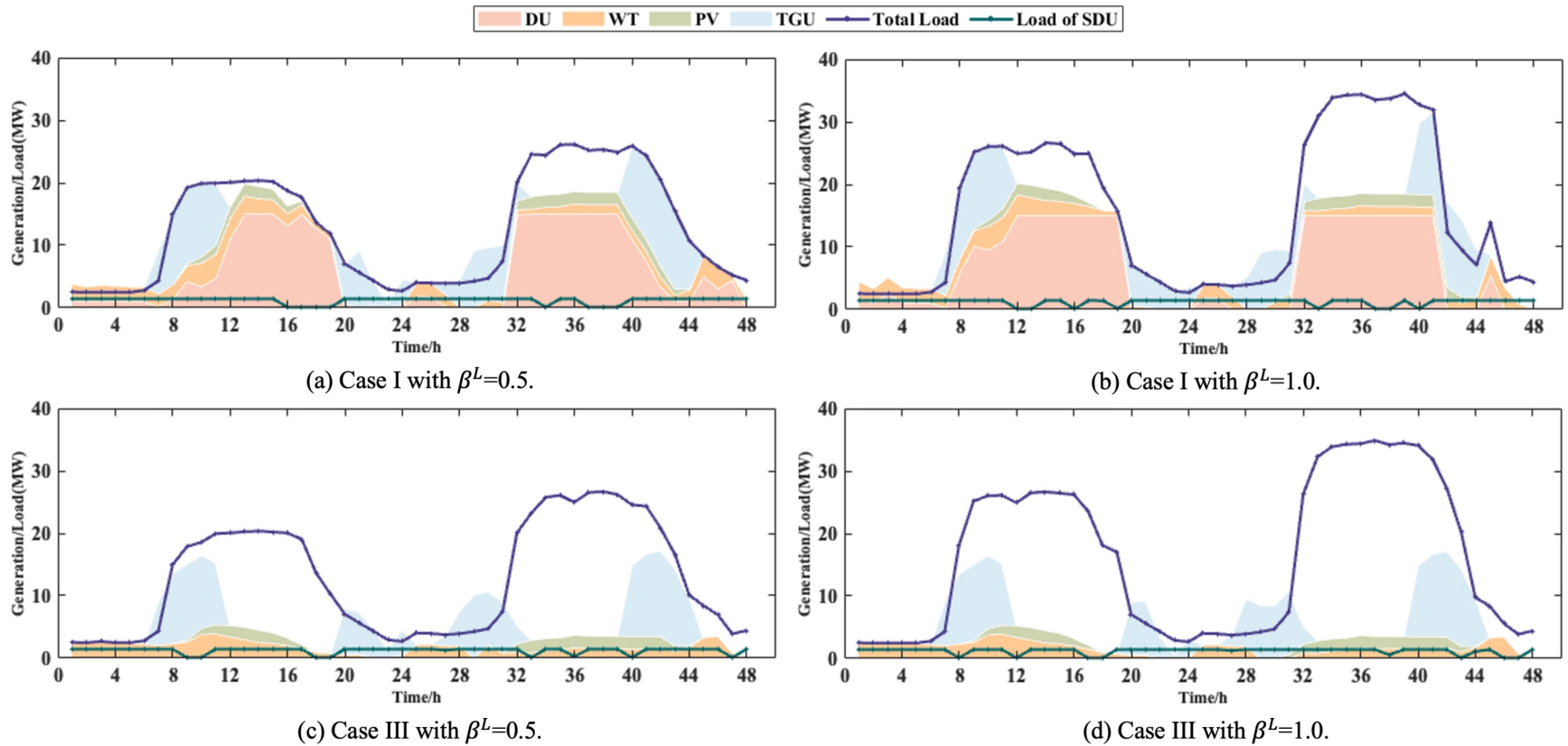
Case #	Installed Capacity (MW)				C^{inv} (M \$)
	DU	NDU	ESS	TPG	
I	15	6	5	14	3.314
II	15	6	-	14	2.834
III	-	6	6	14	2.544
IV	-	6	-	14	1.824
V	15	-	6	14	2.774
VI	15	6	3	-	2.030



- More LS when β^D increases.
- More LS in cases without DU.
- Relatively less LS when adding ESS, but not significant.

Case Studies

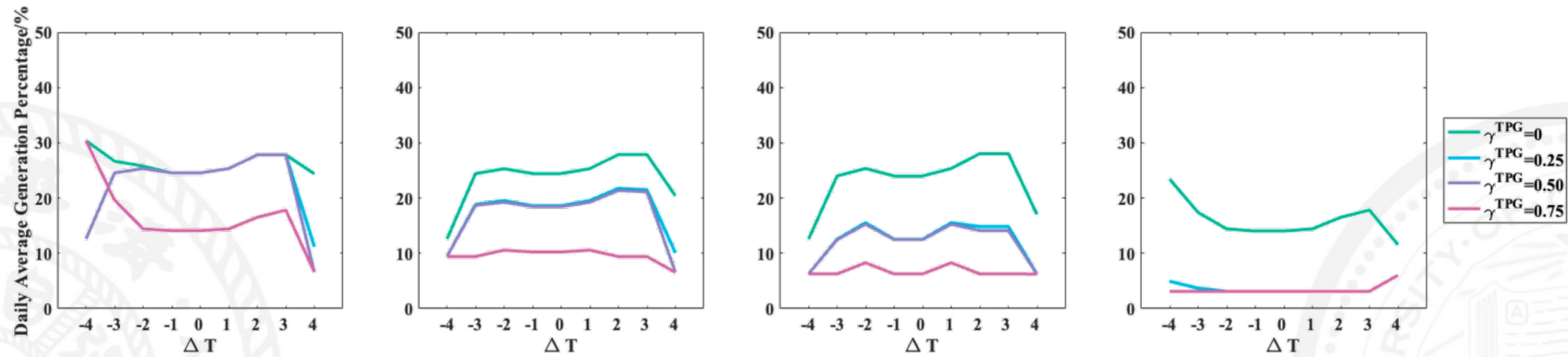
Analysis of Uncertainty of Load Demand



- SDU mainly works in the early morning or late evening, or at noon when PV generation is abundant.

Case Studies

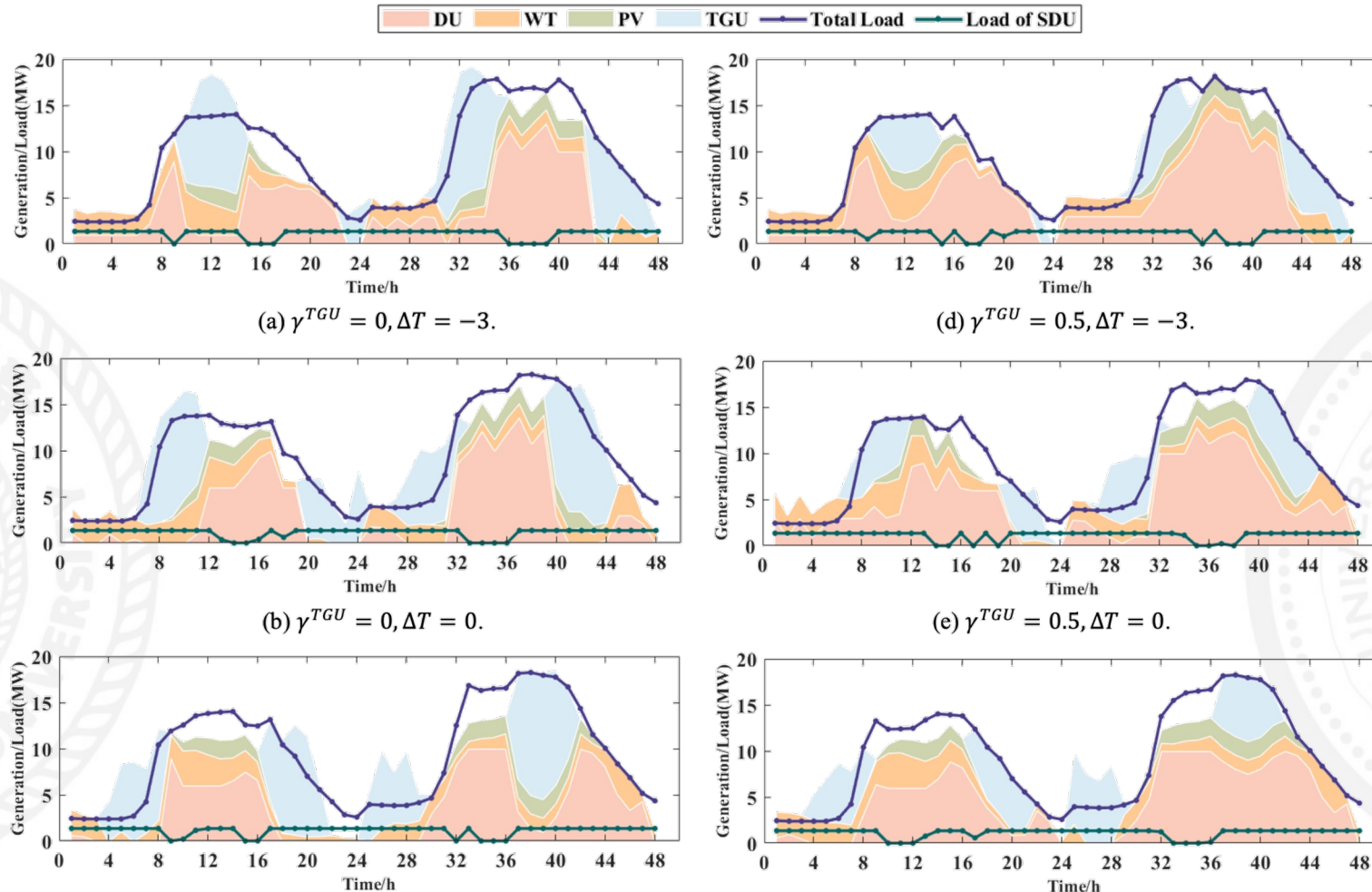
Analysis of Uncertainty of Tidal Power Generation



- Higher γ^{TPG} and β^{TPG} lead to lower generation percentage from TPG.

Case Studies

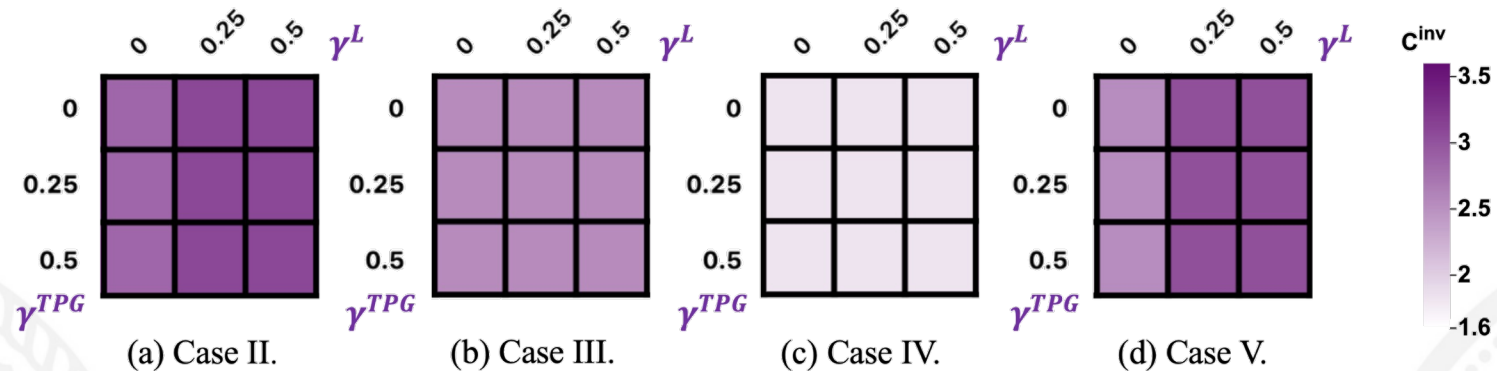
Analysis of Uncertainty of Tidal Power Generation



- Higher daily average generation percentage when ΔT moves the tidal levels to favorable time.

Case Studies

Analysis of Uncertainty of Load Demand and Tidal Power Generation



INVESTMENT DECISIONS AND COSTS UNDER DIFFERENT DER CONFIGURATIONS WITH $\gamma^L = \gamma^{TPG} = 0.5$

Case #	Installed Capacity (MW)				C^{inv} (M \$)
	DU	NDU	ESS	TPG	
II	21	6	-	14	3.098
III	-	6	6	14	2.544
IV	-	6	-	14	1.824
V	21	-	6	14	3.038
VI	21	6	3	-	2.294

- Both uncertainty budget coefficients give rise to C^{inv} , while γ^L contributes more compared with γ^{TPG} .



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04

Conclusions and Future Works





Conclusions and Future Works

- In this paper, a **two-stage robust planning model** for offshore microgrid incorporated with modeling of **tidal power generation** and **seawater desalination** is proposed. The **uncertainties of load demand and tidal power generation** are both modeled and considered. The planning model is solved with the **C&CG** algorithm. Case studies verify the model's effectiveness and analyze the robustness of different DER configuration strategies. Besides, effects of uncertainties are also analyzed.
- The simulation results show the function of energy storage systems in peak shaving, helping to move the load peak and smoothen the load curve. The outputs from tidal generators and PVs show complementarity due to the renewable energy sources' natural characteristics, and the power consumption of seawater desalination units tends to align with the outputs of renewable energy sources. Thus, the combination of them is a valuable part of offshore microgrid planning.
- The future work will focus on considering the connection between **networked microgrids** and developing **accelerating solving algorithm** for the model.



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Thanks for listening!

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